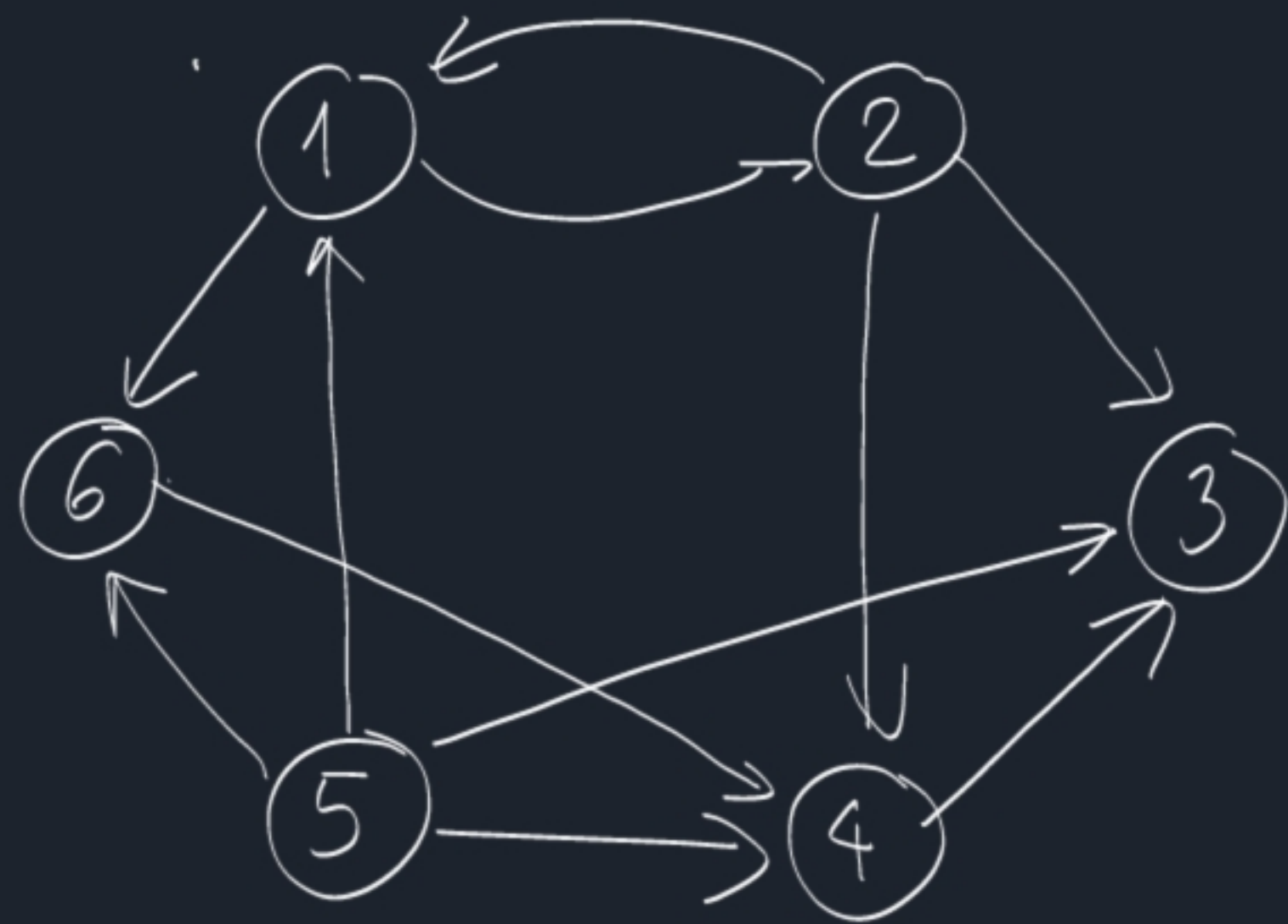




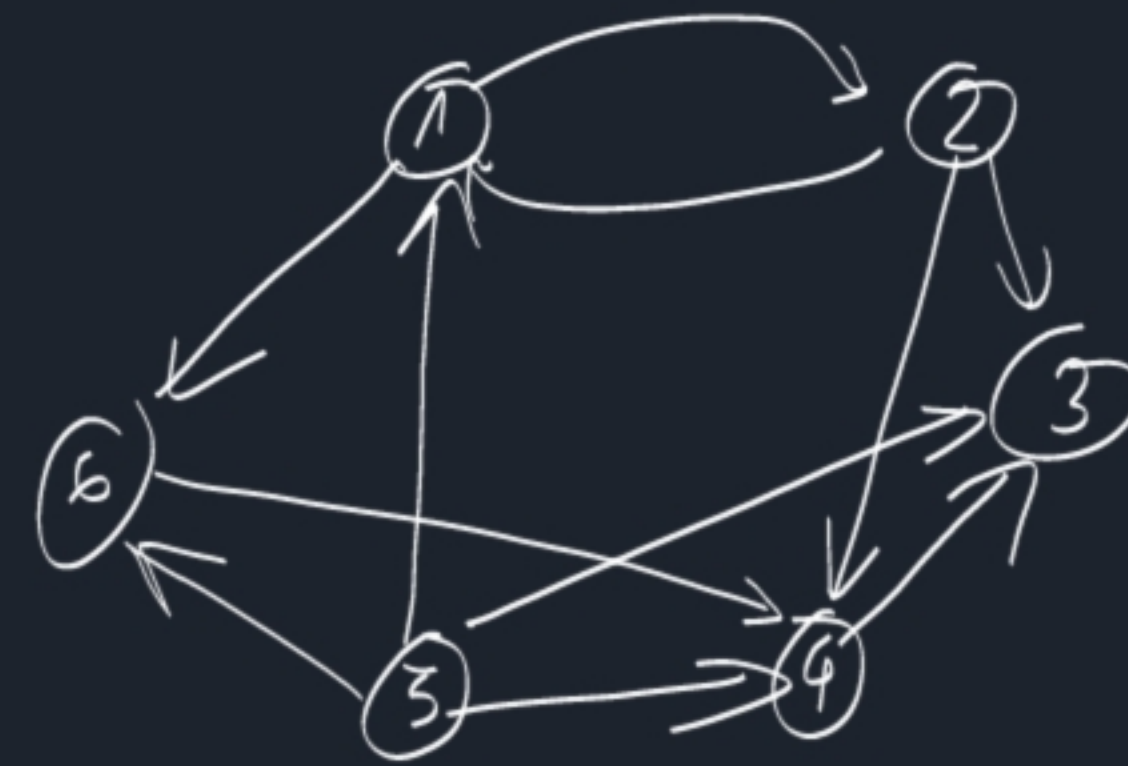
$i \rightsquigarrow i$

$i \rightsquigarrow 5$

	1	2	3	4	5	6
1	1	1	1	1	0	1
2	1	1	1	1	0	1
3	0	0	1	0	0	0
4	0	0	1	1	0	0
5	1	1	1	1	1	1
6	0	0	1	1	0	1

Algorithm to determine transitive closure

① run BFS for every vertex a source vertex

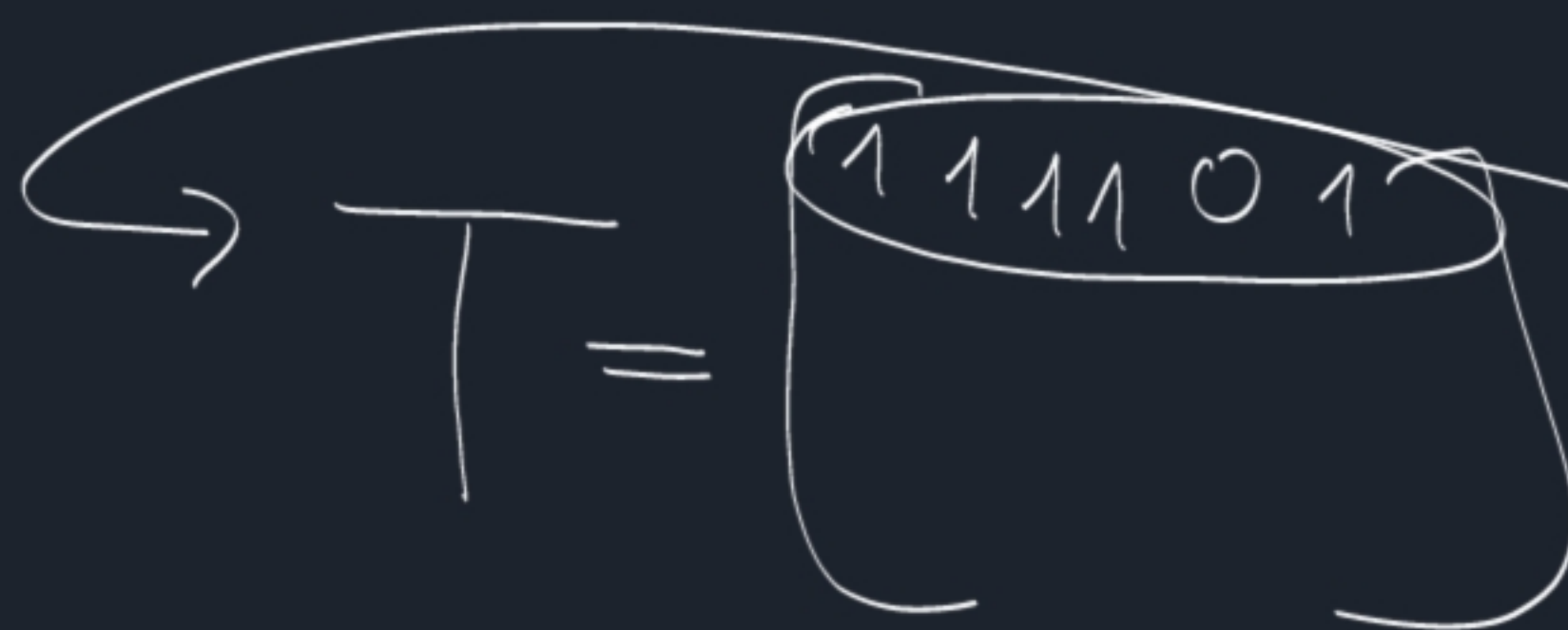


BFS_TC	
$\forall u \in G.V$	
$d(u) := \infty; \pi(u) := \emptyset$	
$color(u) := white$	
$d(s) := 0; color(s) := grey$	
$Q : Queue; Q.add(s)$	
$\neg Q.isEmpty()$	
$u := Q.rem()$	
$\forall v \in G.A(u)$	
$d(v) := d(u) + 1$	SKIP
$\pi(v) := u$	
$color(v) := grey$	
$Q.add(v)$	
$color(u) := black$	

TransitiveClosure($G : \mathcal{G}$) : $\mathbb{B}[n, n]$	
$i = 1$ to n	
BFS_TC(G, i)	
$j = 1$ to n	
$T[i, j] := 1$	$T[i, j] := 0$

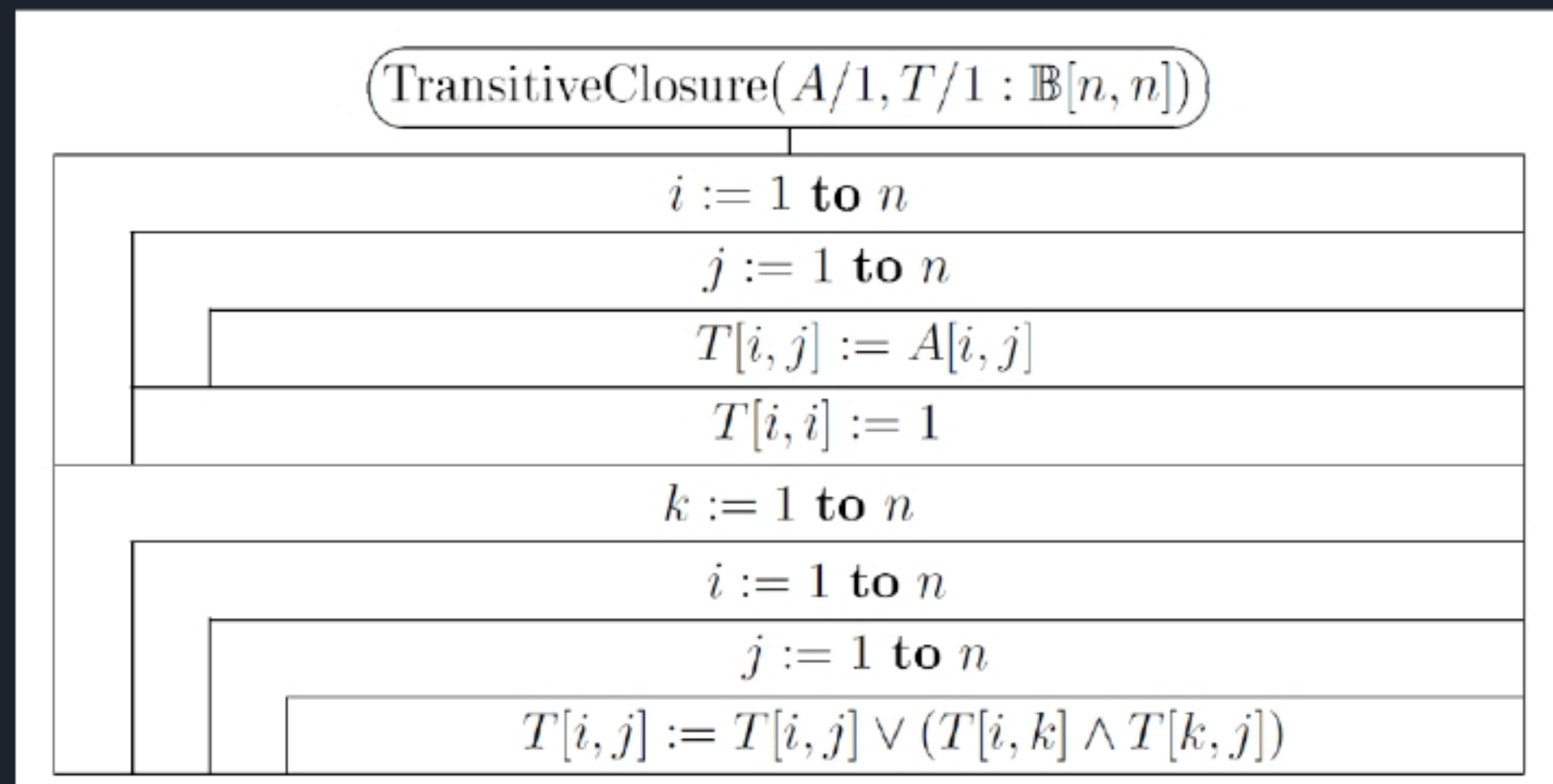
BFS_TC($G, 1$)

1	2	3	4	5	6	Q	vertices exp.
w	w	w	w	w	w	<>	—
g	w	w	w	w	w	<1>	—
g					g	<2,6>	1
	g	g				<6,3,4>	2
						<3,4>	6
						<4>	3
g	g	g	g	w	g	<>	4

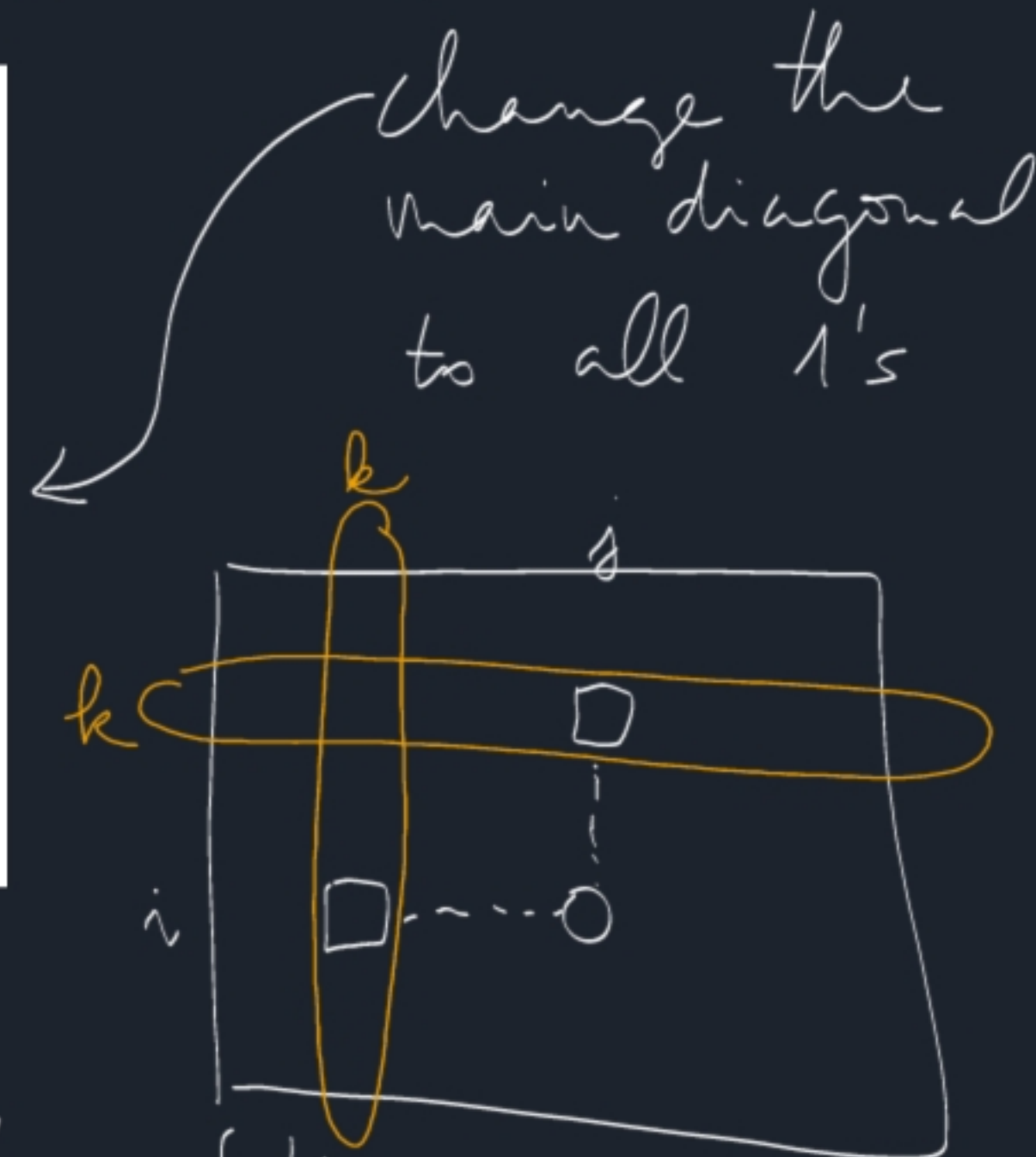


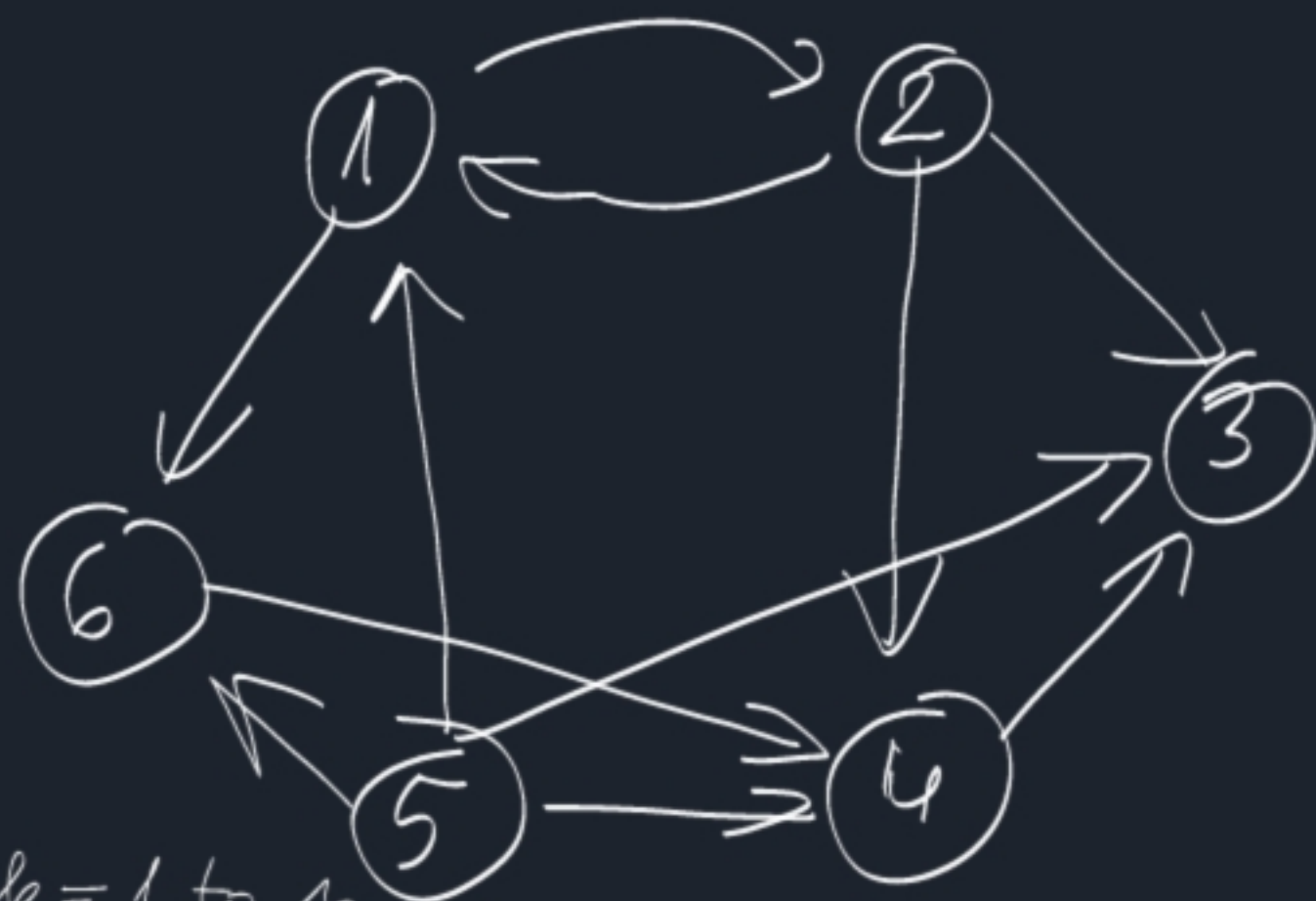
FINAL
VALUES

② Dynamic Programming type algorithm



↪ if $T[i, j] = 1 \Rightarrow$ leave it
if $T[i, j] = 0 \Rightarrow$ if both of the corresponding elements in the highlighted row and column are 1's, then it changes to 1 else it stays 0





$k=1$ to n

$i=1$ to n
 $j=1$ to n
 $T[i,j] = T[i,j] \vee (T[i,k] \wedge T[k,j])$

$$T[i,j] = T[i,j] \vee (T[i,2] \wedge T[2,j])$$

$A =$

0	1	0	0	0	1
1	0	1	1	0	0
0	0	0	0	0	0
0	0	1	0	0	0
1	0	1	1	0	1
0	0	0	1	0	0

$T^{(0)} =$

1	1	0	0	0	1
1	1	1	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
1	0	1	1	1	1
0	0	0	1	0	1

$$T[1,3] \\ T[1,2] \wedge T[2,3]$$

$k=1$

$T^{(1)} =$

1	1	0	0	0	1
1	1	1	1	0	1
0	0	1	0	0	0
0	0	1	1	0	0
1	1	1	1	1	1
0	0	0	1	0	1

$k=2$

$T^{(2)} =$

1	1	1	1	0	1
1	1	1	1	0	1
0	0	1	0	0	0
0	0	1	1	0	0
1	1	1	1	1	1
0	0	0	1	0	1

$k=3$

$$T^{(3)} = T^{(2)}$$

Depth-first search

DFS

DF Visit

